



THÈSE DE DOCTORAT
DE L'UNIVERSITÉ PSL

Préparée à MINES Paris

**Advanced Hemodynamic Modelling and Simulation of Flow
Diverters in the Treatment of Intracranial Aneurysms**

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Le Date de soutenance

École doctorale n°364

**Sciences Fondamentales et
Appliquées**

Spécialité

**Mathématiques Numériques,
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Dedication

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Chapter 1

Introduction

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Abstract

Intracranial aneurysms are complex malformations of the cerebral arteries that remain poorly understood despite their high prevalence. This engineering-focused thesis aims to advance our understanding of these conditions by modeling them in a virtual environment. The first chapter provides a comprehensive overview of aneurysms from a biomedical perspective, summarizing key literature on their formation and growth. We discuss how fluid dynamics simulations, which offer a non-invasive method for studying blood flow associated with these formations, complement findings from autopsies and surgical examinations. Despite their promise, fluid dynamics simulations have not yet resolved many open questions in the field. This thesis specifically addresses two critical aspects: (i) the influence of domain size on aneurysm flow and (ii) the effects of flow diverter stents, which, despite occasional failures, have been instrumental in treating previously untreatable aneurysms.

Resumé

Les anévrismes intracrâniens sont des malformations complexes des artères cérébrales qui restent mal comprises malgré leur forte prévalence. Cette thèse, axée sur l'ingénierie, vise à approfondir notre compréhension de ces conditions en les modélisant dans un environnement virtuel. Le premier chapitre offre un aperçu complet des anévrismes du point de vue biomédical, en résumant les principales publications sur leur formation et leur croissance. Nous discutons de la manière dont les simulations de dynamique des fluides, qui offrent une méthode non invasive pour étudier le flux sanguin associé à ces formations, complètent les résultats des autopsies et des examens chirurgicaux. Bien que prometteuses, les simulations de dynamique des fluides n'ont pas encore répondu à de nombreuses questions ouvertes dans le domaine. Cette thèse se concentre spécifiquement sur deux aspects cruciaux: (i) l'influence de la taille du domaine sur le flux des anévrismes et (ii) les effets des stents à déviation de flux, qui, malgré des échecs occasionnels, ont été essentiels pour traiter des anévrismes auparavant incurables.

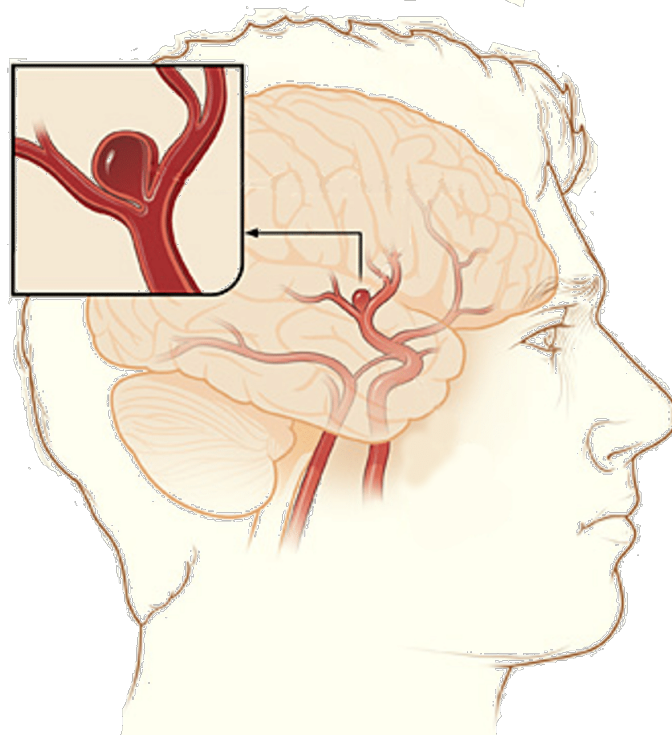
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2 **1.1 Intracranial Aneurysms**3 **1.1.1 Pathology**

4 Intracranial aneurysms (IAs) are abnormal dilations of the cerebral arteries found
5 in an estimated 3.2% [1, 2] to 4% [3] of the adult population. They are often asymp-

6 tomatic and discovered incidentally during imaging studies conducted for other rea-
7 sons. When IAs are symptomatic, patients may experience headaches, visual dis-
8 turbances, and cranial nerve palsies caused by the compression of surrounding brain
9 tissue. The rupture of an IA, which can occur independently of its size and symp-
10 toms, leads to a Subarachnoid Hemorrhage (SAH), a severe condition with a 50%
11 mortality rate [3].

12 During a SAH, blood spills into the subarachnoid space—the area between the arach-
13 noid membrane and the pia mater surrounding the brain. This presence of blood
14 sets off a chain of events, commonly leading to insufficient perfusion¹ of the brain.
15 This deficit in blood flow causes neurological damage, with less than 25% of patients
16 surviving unharmed [4], highlighting the severity of the condition. Fortunately, IA
17 rupture and subsequent SAH are rare events. In fact, 50–80% of IAs are estimated
18 to never rupture [1].



*Figure 1.1: Schematic view of an intracranial aneurysm. This formation resembles a sac-
cular aneurysm found at the middle cerebral bifurcation. Image obtained from
the National Institute of Health for Neurological Disorders and Stroke².*

¹Perfusion: The passage of blood through vessels or other natural channels in an organ or tissue.

1.1.2 Epidemiology

The root causes of intracranial aneurysms are still not thoroughly understood, although many studies consider alterations in local blood flow, genetics, and other environmental factors to be responsible. One step towards understanding the root cause of the disease has been through epidemiology, *i.e.* tracing who has IA and under which circumstances. When discussing Prevalence Ratios (PR), studies often fix an overall PR and then state how a characteristic, *e.g.* gender or age, modifies it multiplicatively. The base value, as previously mentioned, is around 3.2% for a 50-year-old person [2]. Thus, $PR = 2$ would indicate a two-fold likelihood, or a 6.4% prevalence probability.

One of the first apparent observations is that IA is not equally present across different ages and genders, with women being approximately 1.5 to 3 times more likely to have IA than men. Gender not only influences the presence of the disease but where IA spawn, although clear causes have not yet been established [3, 5]. IA incidence is relatively provenance-independent, *i.e.* unruptured IA are roughly equally present in different countries. However, rupture rates vary, as seen in Finland and Japan, where SAH is 2 to 3 times more likely [6]. Genetics also seem to play a role in the PR. The presence of an IA in a direct family member raises prevalence to 9.5%, approximately three times the base occurrence rate. Other inherited conditions, such as Autosomal Dominant Polycystic Kidney Disease, are known to raise the risks ($PR = 6.9$) [2].

As with many other pathologies, smoking increases the risk of IA ($PR = 2.2\text{--}3.9$) and SAH [7–9]. Atherosclerosis³ and hypertension, often accompanying unhealthy habits, generally alter hemodynamics⁴ and vascular health. Atherosclerosis causes plaque formation in vessels, compromising blood flow. Similar depositions have been found in IA, creating a link between both diseases. Hypertension, as the reader may know, implies increased blood pressure in the cardiovascular system, surpassing 140 mmHg at systole and 90 mmHg at diastole. The increased pressure exerted by the heart raises peak flow velocities and pulsatility of arterial flow. An extended exposure to these aberrant conditions is thought to unsettle the balance in the regeneration process of arterial walls, increasing the chances of developing aneurysms [3, 6, 7].

³Atherosclerosis: Disease of the arteries characterized by the deposition of fatty material on their inner walls.

⁴Hemodynamics: The study of blood flow and the forces involved in its circulation through the cardiovascular system.

1.1.3 Pathogenesis

Healthy arteries

A healthy brain artery consists of multiple layers, each with specific structural and functional roles. The innermost layer, known as the endothelium, is a single layer of endothelial cells responsible for biochemical signaling, metabolic exchange, and other regulatory processes [10, 11]. This layer perceives tensile stresses and pressure fluctuations, initiating remodeling cascades to determine the development of subsequent layers. Following the endothelium is the basement membrane and the Internal Elastic Lamina (IEL), a fenestrated layer of elastin fibers. The IEL is crucial for maintaining the elasticity and structural integrity of the vessel wall, especially at arterial bifurcations. In healthy tissue, this layer is well-preserved, ensuring proper function and resilience against stress [7, 10]. The tunica media, the muscular layer of the artery, is primarily composed of smooth muscle cells. These cells regulate blood flow distribution through their ability to contract and relax, a function particularly important in the brain, where arterioles lack the regulatory components found in other body regions. This layer provides the necessary force to manage cerebral perfusion effectively and is particularly well developed in major brain arteries compared to other parts of the body [10]. The outermost layer, the tunica adventitia, consists of collagenous and elastic fibers. These are produced and decomposed by cells called fibroblasts, who reside in this last layer. In brain arteries, the adventitia is sparser and contains a lower proportion of elastic fibers compared to extracranial arteries. Instead of connective tissue, brain arteries are mostly embedded in cerebrospinal fluid, a clear, colorless body liquid found within the tissue that surrounds the brain and spinal cord.

Initiation

As previously stated, initiation is thought to be caused by aberrant hemodynamics acting on weakened arteries. A common site for aneurysms are bifurcations [5]. At these locations, the jet from the parent vessel impacts on a small area, inducing elevated levels of shear stress onto the IEL and the endothelium. By shearing, we understand formally the Wall Shear Stress (WSS), an expression of the friction perceived by an object in contact with a flowing viscous fluid. In zones of high WSS and a positive gradient, *i.e.* a progressive growth of the WSS in flow direction, the arterial wall is observed to lose the IEL, causing an initial dent in the vessel [7, 12]. Extended exposure to these hemodynamics conditions can dilate the lesion further (Fig. 1.2, b, c). During this process, tissue cells in and around the artery attempt to adapt its structure to withstand the strain caused by the flow. An example for this process, can be seen in Fig. 1.2, where one can observe an IA forming on the anterior communicating artery. This particular case has been chosen by the authors

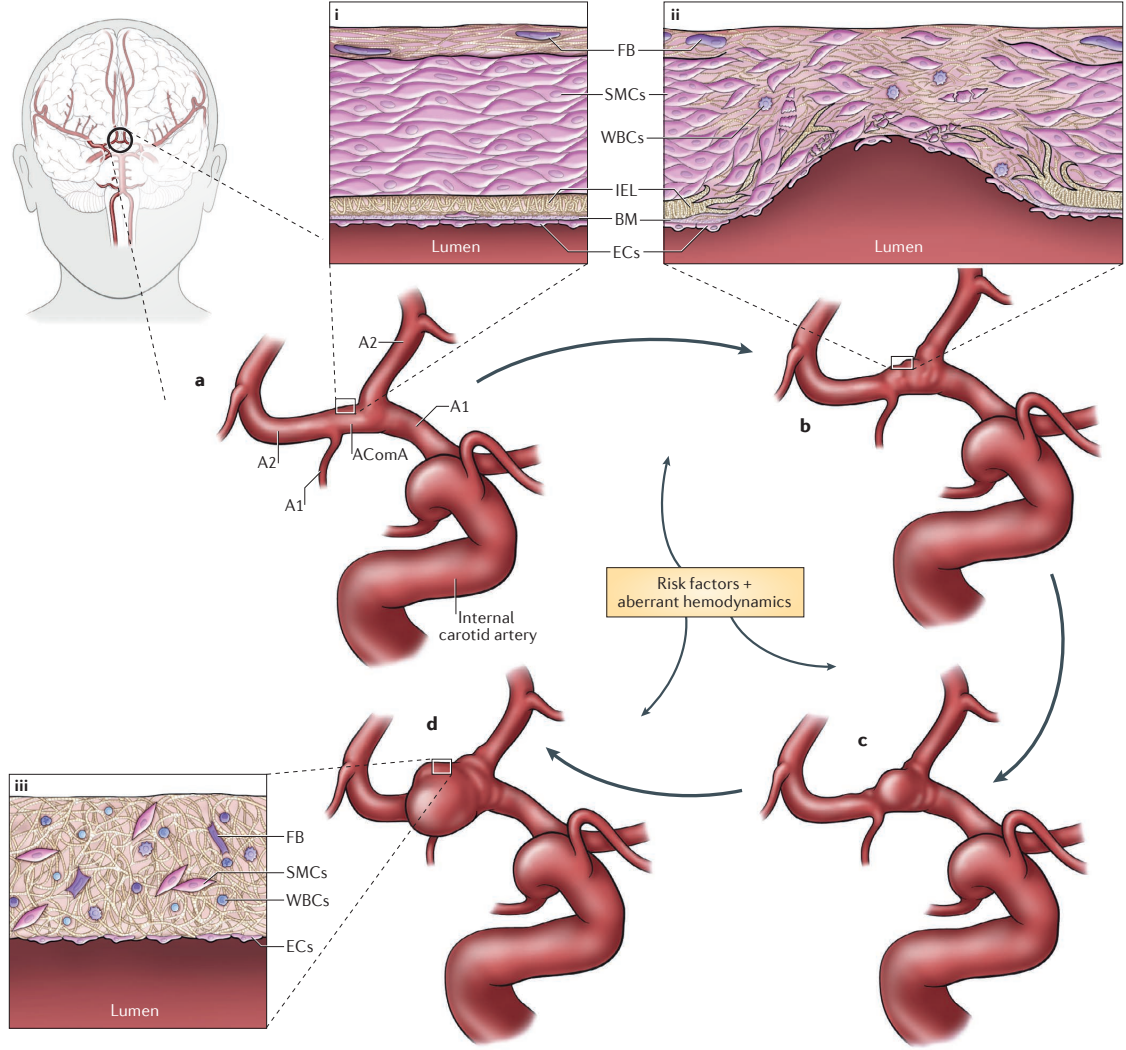


Figure 1.2: Pathogenesis of intracranial aneurysms. Subfigure **a** displays a Anterior Communicating Artery (ACoM) with a healthy wall **i**. In **b** and **c** an aneurysm develops and the vessel wall degrades. Cross section **ii** show a leaky endothelium that permits White Blood Cells (WBCs) to enter the arterial wall. The Smooth Muscle Cells (SMCs) and the Internal Elastic Lamina (IEL) are damaged and disorganized in the bulge area. In **d**, the formation has stabilized. Its tissue (**iii**) mostly contains disaligned fibers and few SMCs. The Figures were taken from Etminan et al. [7]. All rights over the image are reserved to the original authors.

89 of the figure [7] to highlight the growth of IA conditioned by an underdeveloped
 90 side branch. The small vessel left of the aneurysm is usually larger and carries

flow from the other internal carotid artery to the anterior cerebral arteries. The vessels of both brain sides are thereby connected in a circle which helps balancing out the flow, called the Circle of Willis. When some of the vessels in this circle are underdeveloped, as it is here the case, flow velocities can be elevated and lead to an increased prevalence of IA [13]. Note that a detailed anatomical description of the brain arteries will be developed in Chapter 3.

Growth

After the aneurysm initiation, there are three possible scenarios occurring. In the first and most favourable one, the growth stops and homeostasis⁵ establishes. IAs in homeostasis are statistically 30 times less likely to rupture than growing ones [1, 14]. Medical follow-ups are nevertheless carried out to ensure that the formation is stable. This precaution is important due to the particular growth behaviour of IA. As described by Etminan and Rinkel [7], IA rarely go through a continuous change. Instead, growth takes place during short periods separated by long intervals of apparent stasis (see Fig. 1.3 i). The reasons for this rather unintuitive behaviour have, so far, been mostly addressed through hemodynamics studies. There, researches stumble upon seemingly contradicting theories. Studies on one side have highlighted high WSS to be related to the rupture of IA, particularly in the anterior communication [15]. Others [16] found low and oscillatory shearing in growing or ruptured formations.

Given these clashing principles, Meng *et al.* [12] formulated a well-known hypothesis joining both pathways. Once an initial bulge is created through high shearing forces, one of two courses is possible. In the first case, the shearing remains high despite the tissue deformation. Smooth muscle cells start generating inflammatory agents that trigger the remodeling of the arterial wall. In healthy tissue, this remodeling adjusts the structure of the vessel to cope with increased stress. In diseased tissue, it results in an accelerated cell apoptosis⁶ without the desired reinforcement, leading to a hypocellular aneurysm. The thin, sparsely populated walls of such an aneurysm type can be seen on Fig. 1.4 marked with an A.

In the second scenario, the mean WSS levels drop as a consequence of the aneurysms ballooning, as displayed in Fig. 1.3. Frequently, the drop in WSS is accompanied by oscillatory shearing, which are common in atherosclerosis [12]. The oscillatory flow, formally quantified through the Oscillatory Shear Index (OSI), disrupts the junctions between endothelial cells and elicits a pro-inflammatory behaviour [11]. The first consequence is increased permeability, allowing white blood cells to traverse the endothelium and penetrate into the artery (see Fig. 1.3, iii). There, they spark

⁵Homeostasis: State of equilibrium in a physiological process.

⁶Apoptosis: The process of programmed cell death.

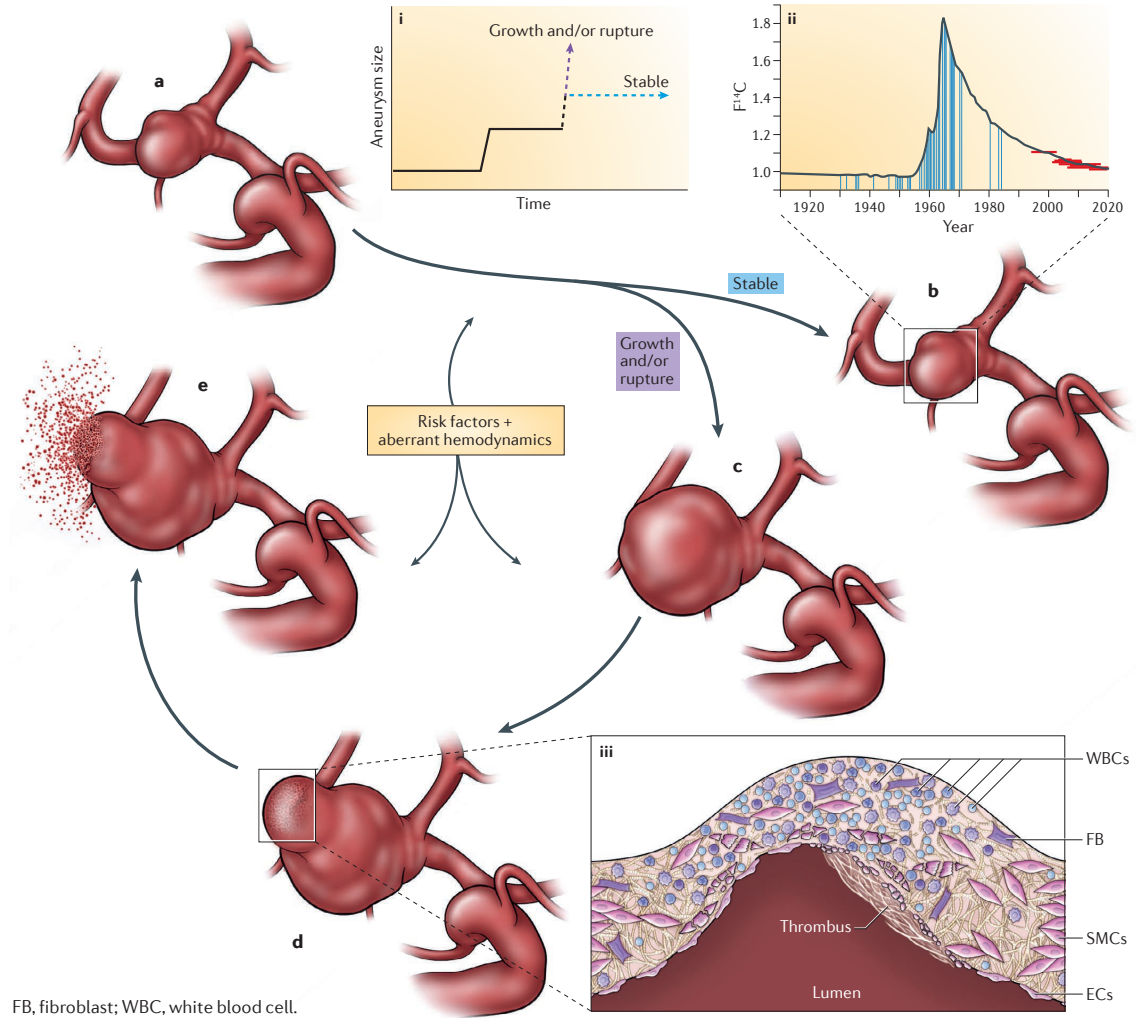


Figure 1.3: Evolution of an intracranial aneurysm. A stable pathway, is shown in **b**. Carbon dating (**ii**) has shown that IA of the people born at the years marked in blue are formed in recent years (red), rather than in their infancy. Growth, shown in **c** and **d**, is characterized by periods of instantaneous enlargement and stasis (**i**) that can lead to rupture (**e**). The aneurysm tissue before rupture contains weak zones (**iii**) where destructive remodelling has led to an absence of SMC and fibers to hold the structural load [7]. Accompanying this phenomenon are layers of thrombi forming on the inner IA surface due to the exposure of medial layer. The Figures were taken from Etminan et al. [7]. All rights over the image are reserved to the original authors.

127 inflammation of the tissue, much like in a lesion, degrading the extracellular matrix
 128 and promoting the accumulation of plaque and lipoproteins [12]. As a result, the
 129 IA walls become thicker and attain a white appearance as seen in aneurysm D of

130 Fig. 1.4. Both remodeling pathways can coexist, with some regions of an IA being
131 subject to different external stresses. A possible outcome of a combined remodelling
132 could be reflected on aneurysms like B and C in Fig. 1.4 [12], showing an inhomoge-
133 neous wall constitution. These mechanisms may lead to different growth dynamics.
134 In the most worrying case, the IA develops daughter sacs or blebs. These further
135 compromise the structural integrity of the IA, forcing clinicians in most cases to
operate [12].

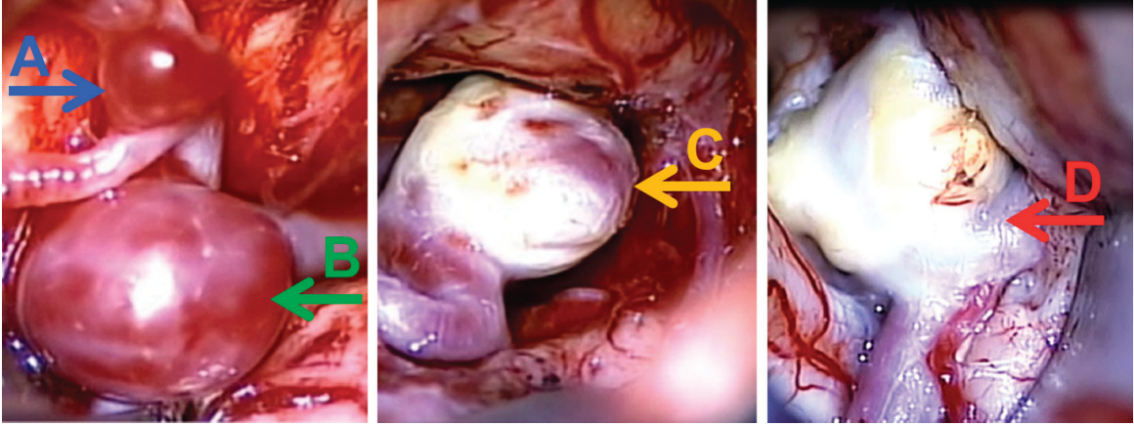


Figure 1.4: Images of unruptured IA taken during surgical interventions. The inscribed letters note IA with different wall constitutions. A is attributed to a high WSS related IA with a translucent hypocellular wall. B and C are mixed IA with hypocellular and atherosclerotic patches. D is the consequence of low and oscillatory WSS and features a completely white appearance due to the presence of plaque. Taken from [12].

136

137 1.1.4 Treatment

138 Neuroradiologists wage benefits and risks of treatment against the threat of an
139 aneurysm rupture. Indicators for a probable rupture are as mentioned growth and
140 the presence of daughter sacs. The size ponders also into the decision, since large
141 (11–24mm) and giant aneurysms (≥ 25 mm) are more likely to be symptomatic
142 and thus need to be treated. In general, the correlation between size and rupture
143 risk remains controversial. Until recently, the rupture risk compared to medium (7–
144 10mm) and small IA (< 7 mm) was assumed to be smaller [17]. Studies analyzing
145 ruptured IAs concluded that if size was mostly used for assessment, 78% of ruptured
146 cases would not have been classified as at risk [18]. It is also important to consider
147 that most IAs belong to the “small” category and that conditional dependence is
148 crucial when defining risk metrics. The age of the patient also plays a decisive

149 role. While the yearly rupture risk is higher for older patients (≥ 70 years) [17], the
 150 cumulative risk for younger individuals is larger due to their longer life expectancy.

151 The list of considerations that clinicians must take into account when deciding on
 152 a treatment is long and ultimately depends on the patients will. The motivation
 153 behind these careful considerations is that any intervention entails a non-negligible
 154 risk for the patient, and furthermore, not all treatments meet their intended purpose.
 155 Bijlenga *et al.* [17] state that a conservative approach is justified in patients for whom
 156 the risk of treatment complications is higher than the 5-year rupture risk. In regards
 157 to this rule, we have seen that rupture risks are inhomogeneous and depend on many
 158 (unknown) factors. The risk of treatment is likewise hard to estimate and usually
 159 depends on the employed strategy. To put the reader into context we provide a
 160 short overview of the history of IA treatment.

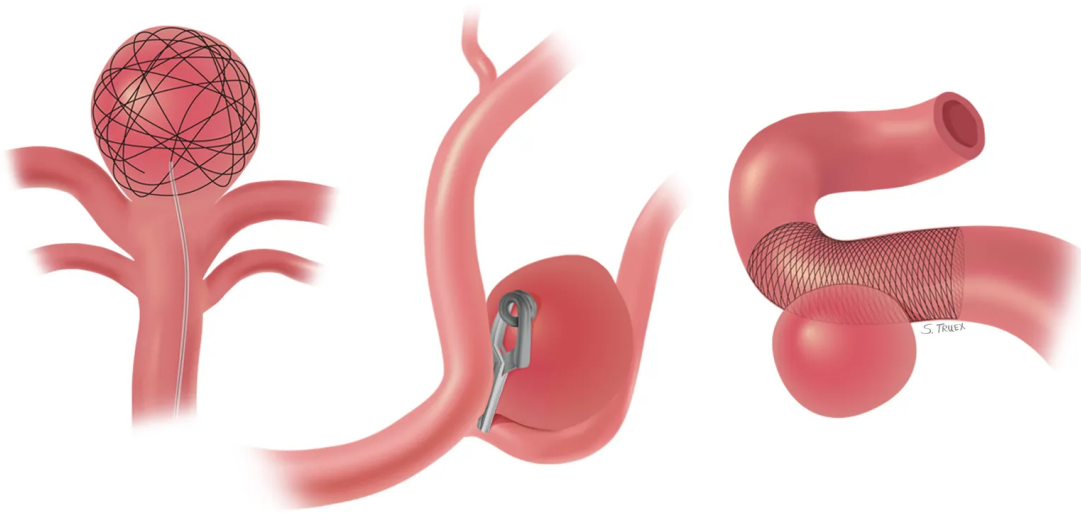


Figure 1.5: Schematic overview of the different treatment options for intracranial aneurysms (catheters and wires not in scale). From left to right: Endovascular coiling, surgical clipping, and stenting. Taken from the UTSouthwestern Medical Center⁷.

161 Since the 19th century, doctors have attempted to treat IA with various methods,
 162 primarily by cutting the flow in the parent artery and inducing thrombosis of the
 163 aneurysm. Unsurprisingly, these initial operations often resulted in the immediate
 164 death of the patient [19]. This was not necessarily due to the chosen method but
 165 rather the unhygienic conditions under which these early interventions were carried
 166 out. Nowadays, the embolization⁸ of a parent vessel can be considered in cases where

⁸Embolization: Blockage of a vessel by natural or surgical means, such as coils.

167 other treatments are not applicable, provided sufficient peripheral flow reaches the
168 distal sections past the aneurysms.

169 In 1937, Walter Edward Dandy performed the first surgical clipping operation [20].
170 In this procedure, after opening the patient's skull, the aneurysm neck is ligated
171 with a metallic clip, isolating the bulge from blood flow and preventing rupture
172 (see Fig. 1.5). At the time, this method revolutionized IA treatment and remains
173 one of the most common treatments today in some parts of the world. However,
174 its disadvantages are considerable. First, open brain surgery is risky due to the
175 delicate environment and the risk of infections. Secondly, the use of a clip is only
176 feasible when the aneurysm neck is sufficiently narrow to maintain the structure,
177 substantially reducing the applicability. Other secondary motivations, such as the
178 increased costs of patient care during the longer recovery time make this approach
179 less attractive in Western countries.

180 The introduction of endovascular techniques in the 1990s led to a shift in the treat-
181 ment paradigm. Unlike surgical approaches requiring a craniotomy⁹, treatments
182 performed through the catheterized parent vessel bear a considerably smaller risk
183 of complications [4]. More precisely, during endovascular treatment, the patient is
184 put under general anesthesia, and a catheter¹⁰ is inserted into the arterial system,
185 commonly through the femoral artery. Apart from being a large, accessible vessel,
186 the femoral artery is situated far from the brain, reducing the risk of critical in-
187 fections and shortening recovery times. The catheter is then navigated upstream
188 through the aorta into either the carotid or vertebral arteries until reaching the
189 aneurysm. In the first proposed endovascular treatment, coiling, detachable plat-
190 inum wires are inserted into the aneurysm sac, slowing down the intrasaccular flow
191 to a point where a stable thrombus can form (see the left of Fig. 1.5). The thrombus
192 then seals the aneurysm from the parent vessel, preventing rupture. This method,
193 when applicable, is considered the gold standard in IA treatment, even after rup-
194 ture [21]. Unfortunately, its applicability is not universally guaranteed. Coils need
195 to be safely introduced into the aneurysmal sac without the risk of detachment,
196 as loose wires in the bloodstream would have fatal consequences. Therefore, the
197 aneurysm neck must be sufficiently narrow to maintain the structure, excluding
198 wide-necked and fusiform¹¹ aneurysms from treatment. The latter IA phenotype,
199 with a 10% prevalence, is substantially less frequent than its saccular counterpart
200 and less well-studied [22]. To extend intrasaccular treatment, new devices have been
201 proposed, such as the Woven EndoBridge (WEB). This device is built with a set of
202 interwoven metallic wires that are fixed at both ends, forming a foldable cylindrical

⁹Craniotomy: The surgical removal of part of the bone from the skull to expose the brain.

¹⁰Catheter: A flexible tube inserted through a narrow opening into a body cavity.

¹¹Fusiform: A spindle-like shape similar to one of a Zeppelin.

device. It is deployed, similarly to coils, through a transluminal catheter into the aneurysm sac. The design of this device makes it more suited for aneurysms situated at vessel bifurcations [23]. Both of coils and the WEB device have a potential risk of aneurysm recurrence and neck remnants, in which case a second intervention is needed.

Despite these numerous approaches, many IAs still remained outside the treatability range and required supportive means. One tool that facilitates coil deployment are braided stents. Their normal intended use is, similarly to coronary stents, widening stenosis and improving the blood flow. Here however, the stent is deployed over the neck to provide a scaffold to coils in the sack. While extending the range of applicability, this method still is not enough to handle all untreatable cases [24]. In some instances, clinicians employed stents alone, to avoid perforating the weakened aneurysm tissue while deploying the coils. They observed that a minor flow redirection sometimes sufficed to improve the patient's condition. This led to the development of flow diverter stents, closely related to intracranial stents, whose main distinguishing feature is a high wire density with reduced mechanical strength [25]. These properties follow from using 2–3 times more wires with only half of their thickness approximately [26–28]. Since their approval in the early 2010s, Flow Diverter Stents (FDS) have enabled clinicians to treat complex, fusiform IAs without sacrificing the parent vessel or resorting to surgery. When used, the clinician measures the parent vessel and selects the proximal¹² and distal landing zones. The selected FDS is deployed by advancing a catheter past the aneurysm and unsheathing the device while pulling the catheter back. Once in place, the wires redirect flow away from the bulge, allowing a stable thrombus to form inside the cavity. After a successful intervention, the thrombus is slowly converted into collagen, leading to the reabsorption of the IA. Like all previously mentioned methods, this one also has downsides. Due to the presence of metallic wires in the bloodstream, patients experience an increased risk of blood clot formation inside the artery. These clots can grow, travel downstream, and cause a stroke if not prevented. For this reason, patients carrying stents and FDS are prescribed with (dual) antiplatelet medication for the rest of their lives [25, 27, 28]. Additionally, these devices can lead to complications when covering bifurcating vessels, as they can either clog them or cause a thrombus to form inside them. This clash between potential benefits and risks has led to a heterogeneous use of FDS in the medical community, reflected also in widely scattered success rates across studies [28]. In the latter review study, success rates ranged from 49% to 100%, raising numerous questions about how and under which circumstances these devices are used. It is also clear that the function of FDS is not fully understood by the community, a fact that is surely related to the complexity

¹²Proximal: In the medical context, the side closest to the heart. In that sense, the opposite to distal.

of aneurysms and their formation, growth, and rupture mechanisms.

1.2 Hemodynamics of IA

Numerous questions risen from the IA growth and treatment call out for a deeper analysis beyond statistical outcomes after treatment. As we have seen, the disease and the treatment options depend largely on the hemodynamics. There are several ways of obtaining a picture of the aneurysmal flow: *in-vivo* measurements, *in-vitro* experiments and lastly *in-silico* simulations. We will briefly discuss these below and work on some measurement modalities in chapter 3.

1.2.1 Why Simulations Matter

Patient data is overall the most valuable form of information, since it stems directly from the patients and thus carries the smallest amount of bias. Nevertheless, it comes at costs and is not generally available. First, some measurement techniques, such as Computed Tomography (CT), expose patients to radiation that however harmless it seems, needs to be waged against their possible benefits of the targetted study. This is mostly the case when considering health subjects for reference populations, since they have no medical reason to undergo those checks. Another difficulty is the applicability. Doppler-Ultrasound, for instance, is non-invasive and can measure the flow velocities, but is limited to areas that are not covered by bones-a problematic limitation when considering the brain. Experimental and computational methods come into play at this stage, by attempting to bridge and combine the scarce recordings to obtain the full picture. In-vitro setups can use realistic silicone models in which blood-like substances, such as glycerol, are pumped through in regular pulses. In methods like Particle Image Velocimetry, the fluid is seeded by small fluorescent particles that can be tracked to obtain a 3D map of the flow velocity. This technique has been long used to study the flow in diverse environments and to validate and calibrate simulation models. Compared to simulations, experiments manage to avoid certain modelling steps, such as solving expensive numerical problems and provide valuable insights where the physics are not well understood[29]. There are however two main reasons, why in-vitro experiments are not the go-to method for IA studies. First, experimental setups are time consuming for researchers to carry out, as they require lots of preparation, post-processing and repetitions to ensure the results are reliable. Secondly, results are limited to the setup and the used materials, which can be a problem when trying to generalize the findings. In the case of IA, for instance, blood cannot be used due to its low opacity and thrombogenic behaviour outside a living organism. Similarly to simulations, experiments also rely on knowledge about boundary conditions and material properties, providing no further insight into these

277 complex aspects.

278 1.2.2 Computational Fluid Dynamics

279 Boosted by the improving performances of computing systems, CFD has become
280 a fundamental tool in research and development. In industry, simulations permit
281 testing in parallel arrays of product designs for optimization purposes, saving time
282 and resources for prototyping. In science, CFD can recreate plenty of natural phe-
283 nomena and provides a usefull tool for hypothesis testing. It can also be used to
284 prognose the intensity of catastrophic phenomena, such as tsunamis, supporting the
285 decision-making process. In biomedical applications, CFD are exploited to recreate
286 the hemodynamics in narrow, confined spaces, elucidating in detail the flow patterns
287 following imposed boundary conditions.

288 Broadly speaking, in CFD a fluid domain is considered as an assembly of small, dis-
289 crete elements on which the solution to some governing equations are approximated.
290 Then, the solutions are reassembled from all of the small contributions, providing a
291 macroscopic view of the flow. The dynamics of the system are usually driven by the
292 Navier-Stokes equations, which describe the conservation of momentum, mass, and
293 energy in the case of compressibility. The users of CFD prescribe a set of bound-
294 ary conditions and material properties following the known system characteristics.
295 Naturally, each system is subject to different levels of complexity, like the presence
296 of turbulence, compressibility, or multiphasic flow. These impact the choice of the
297 solver and the computational resources needed to solve the problem in feasible time.
298 Finding a set of suitable simplifications is not an easy task and requires a careful
299 consideration. Therefore, Chapter 2 will be entirely dedicatedn to the description
300 of the governing equations and the numerical methods used in CFD.

301 1.2.3 CFD in Biomechanics

302 Accomplishments

303 So far we have shown the interest of learning more about the hemodynamics of in-
304 tracranial aneurysms and designated CFD as the tool to provide us with the sought
305 insights. This procedure has been already applied to a number of applications in the
306 human body with highly satisfactory outcomes. Cardiovascular research commonly
307 profits from hemodynamics analysis, as for instance in Taylor *et al.*(2013) [30], where
308 a better estimation of the pressure drop following coronary stenosis was found with
309 CFD than with traditional invasive methods. As one of the most prominent exam-
310 ples, this finding has proven that in-silico tests can avoid risks and complications for
311 patients first-hand. Other research groups have successfully related plaque forma-
312 tion to altered hemodynamics across different locations in the arterial system [31].

These findings help understanding how in principal it could be possible to predict the progression of the diseases and help clinical decision making. Many other applications devote their time to simulating and optimizing medical devices, from heart valves [32] to dialysis machines (see [33] and therein). All of these accomplishments came after exhaustive, often multi-centered analysis. This includes iterations of hypotheses and several checks until a validated consensus is found.

Motivation

Computational Fluid Dynamics has proven to be a valuable tool in advancing research and development in various diseases. Intracranial aneurysm research, in particular, has benefited from this discipline, helping to clarify some of the underlying disease mechanisms [12]. However, there remains significant room for improvement and validation of the proposed theories, as discussed earlier. The two primary areas of focus are the study of the pathology and its treatments. Regarding the pathology, advances in imaging and measurement technologies have enabled larger studies that incorporate high-fidelity data with statistical and methodological significance. These advances allow for the identification of correlations that can enhance our understanding of the disease.

The second major research area for CFD and solid mechanics in the context of IA is the optimization of treatment strategies. Beyond the obvious benefits of refining treatment methods, this research also enhances our understanding of the hemodynamic changes that contribute to positive disease outcomes. For “conventional” saccular aneurysms, guidelines already exist for coil packing, stent porosity, and placement [25]. However, less common aneurysms, such as blister-like, fusiform, and giant aneurysms, are not specifically addressed. These types of aneurysms are associated with higher morbidity and mortality rates [34, 35] and have limited treatment options available. Flow diverters are increasingly being used in these cases due to their versatility and ability to conform to the shape of the aneurysm [36]. However, in the absence of established treatment guidelines, clinicians often rely on intuition, sometimes deploying multiple stents, flow-diverter stents, or even add coils to achieve the desired occlusion of the aneurysm. To improve outcomes and define what constitutes an optimal result, large studies could be conducted to closely examine these cases. This would include simulations of fluid dynamics and stent deployment to determine which flow reductions and stent porosities yield the best outcomes.

Current Bottlenecks

Currently, the progress in this field is throttled by two factors, (i) an inconsistent choice of techniques and approximations when carrying out the simulations and (ii) data availability. The first issue was identified by researchers like Steinman,

Berg and Valen-Sendstad [37–39] when coming across conflicting theories about IA growth. They hypothesized that research teams were employing modelling assumptions so different from each other that their results would lead them to other interpretations. They tested this hypothesis by opening computing challenges in which participants would simulate the same cases and report their results. With those, the organizers could analyze the deviations and assess the research question. Berg and Voß *et al.* [40, 41] isolated perhaps the largest source of dissent, the segmentations. Due to the complex and per-definition irregular shape of aneurysms, this task is subject to variability even when carried out by clinicians. In many other cases, they found that the community chooses inconsistently their approximations [42]. Interestingly, when it came to selecting aneurysms that would likely rupture, no clear strategy prevailed over the other.

The reason for this turmoil lies ultimately in the lack of measurement data that forces the use of approximations. Luckily, imaging technology is advancing in recent times introducing more potent Magnetic Resonance Tomographs and Computed Tomography scanners, among others. These will likely improve data availability thereby alleviating data-related issues.

The second factor, data availability, is conditioned by the relatively slow exchange of medical and the less frequently appearing aneurysm phenotypes. Due to the sensitive nature of medical data, it is important to ensure a responsible handling of patient information. Additionally, to obtain a statistically relevant cohort size for studies, researchers are required to interview many medical institutions, repeating the weary process a number of times. As a consequence, less attention has been given to giant and blister-like aneurysms, despite their relatively high propensity to complications, 20–25% [43, 44] and 38% [34], respectively.

1.3 Goal and Structure

We have identified an intrinsic need for researching aneurysms, particularly complex types such as blood-blister and giant aneurysms, to enhance their treatability. To address this, our goal is to develop a high-fidelity framework that simulates aneurysm hemodynamics in both preoperative and postoperative states. Our work will focus on the treatment option of flow diverters, which we will deploy using a new algorithm based on geometrical constraints.

This thesis is structured into four chapters, each detailing the necessary steps in the development process. Chapter 2 revisits fluid dynamics theory, starting from the constitutive equations and progressing to discretized systems. This chapter also includes a discussion on rheology modeling and introduces a novel method in stabilized Finite Elements, which verified using two classic benchmarks from CFD

387 literature.

388 In Chapter 3, we explore the fundamental properties of brain blood flow, with a
389 focus on mesh generation and boundary conditions. Additionally, we examine the
390 influence of the simulated domain size on aneurysm hemodynamics, using three
391 patient-specific arterial systems as case studies.

392 Chapter 4 introduces a geometry-based flow diverter deployment model, providing
393 a robust and efficient tool for predicting device sizes, porosities, and generating wire
394 meshes. The algorithm is thoroughly validated against examples from literature,
395 in-vitro experiments, and an in-vivo case.

396 Finally, in Chapter 5, we apply the developed methods to three patient cases involv-
397 ing ruptured giant, blister-like, and wide-neck aneurysms. The giant aneurysm is
398 studied in relation to delayed rupture phenomena after treatment, including fluid-
399 structure interaction simulations conducted in collaboration with Aurèle Goetz. The
400 other two cases focus on post-rupture treatment, showcasing the changes in hemo-
401 dynamics from an unstented state to the use of two telescopic stents.

References

- [1] R. G. Tawk, T. F. Hasan, C. E. D'Souza, J. B. Peel, and W. D. Freeman, *Mayo Clinic Proceedings* **96**, 1970 (2021).
- [2] M. H. Vlak, A. Algra, R. Brandenburg, and G. J. Rinkel, *The Lancet Neurology* **10**, 626 (2011).
- [3] G. J. Rinkel, M. Djibuti, A. Algra, and J. V. Gijn, *Stroke* **29**, 251 (1998).
- [4] A. J. Molyneux, R. S. Kerr, L.-M. Yu, M. Clarke, M. Sneade, J. A. Yarnold, and P. Sandercock, *The Lancet* **366**, 809– (2005).
- [5] R. D. Brown and J. P. Broderick, *The Lancet Neurology* **13**, 393 (2014).
- [6] N. Etminan, H. S. Chang, K. Hackenberg, N. K. De Rooij, M. D. Vergouwen, G. J. Rinkel, and A. Algra, *JAMA Neurology* **76**, 588 (2019).
- [7] N. Etminan and G. J. Rinkel, *Nature Reviews Neurology* **12**, 699 (2016).
- [8] W. Ji, A. Liu, X. Lv, H. Kang, L. Sun, Y. Li, X. Yang, C. Jiang, and Z. Wu, *Stroke* **47**, 971 (2016).
- [9] B. G. Thompson, R. D. Brown, S. Amin-Hanjani, J. P. Broderick, K. M. Cockroft, E. S. Connolly, G. R. Duckwiler, C. C. Harris, V. J. Howard, S. C. C. Johnston, et al., *Guidelines for the Management of Patients With Unruptured Intracranial Aneurysms: A Guideline for Healthcare Professionals From the American Heart Association/American Stroke Association*, vol. 46 (2015).
- [10] D. S. Liebeskind and L. R. Caplan, *Frontiers of Neurology and Neuroscience* **40**, 1 (2016).
- [11] C. A. Dessalles, C. Leclech, A. Castagnino, and A. I. Barakat, *Communications Biology* **4** (2021).
- [12] H. Meng, V. M. Tutino, J. Xiang, and A. H. Siddiqui, *American Journal of Neuroradiology* **35**, 1254 (2014).
- [13] L. B. Hindenes, T. Ingebrigtsen, J. G. Isaksen, A. K. Håberg, L. H. Johnsen, M. Herder, E. B. Mathiesen, and T. R. Vangberg, *Journal of the Neurological Sciences* **452** (2023).
- [14] W. Brinjikji, Y. Q. Zhu, G. Lanzino, H. J. Cloft, M. H. Murad, Z. Wang, and D. F. Kallmes, *American Journal of Neuroradiology* **37**, 615 (2016).
- [15] M. Castro, C. Putman, M. Sheridan, and J. Cebal, *American Journal of Neuroradiology* **30**, 297 (2009).
- [16] J. Xiang, S. K. Natarajan, M. Tremmel, D. Ma, J. Mocco, L. N. Hopkins, A. H. Siddiqui, E. I. Levy, and H. Meng, *Stroke* **42**, 144 (2011).
- [17] P. Bijlenga, R. Gondar, S. Schilling, S. Morel, S. Hirsch, J. Cuony, M. V. Corniola, F. Perren, D. Rüfenacht, and K. Schaller, *Stroke* **48**, 2105 (2017).

References

- [18] I. Pagiola, C. Mihalea, J. Caroff, L. Ikka, V. Chalumeau, M. Iacobucci, A. Ozanne, S. Gallas, M. Marques, D. Nalli, et al., *Journal of Neuroradiology* **47**, 349 (2020).
- [19] R. R. Smith, Y. N. Zubkov, and Y. Tarassoli, *The History of Aneurysm Surgery* (Springer US, 1994), pp. 1–9, ISBN 9781461395324, URL http://dx.doi.org/10.1007/978-1-4613-9532-4_1.
- [20] D. Laurent, B. Lucke-Wold, O. Leary, M. H. Randall, K. Porche, M. Koch, N. Chalouhi, A. Polifka, and B. L. Hoh, *Neuroscience Insights* **17**, 26331055221117560 (2022), pMID: 35924091.
- [21] S. A. Pineda-Castillo, E. R. Jones, K. A. Laurence, L. R. Thoendel, T. L. Cabaniss, Y. D. Zhao, B. N. Bohnstedt, and C. Lee, *Systematic review and meta-analysis of endovascular therapy effectiveness for unruptured saccular intracranial aneurysms* (2024).
- [22] A. Keedy, *MJM* **9**, 141 (2006).
- [23] C. Kabbasch, L. Goertz, E. Siebert, M. Herzberg, J. Borggreffe, A. Mpotsaris, F. Dorn, and T. Liebig, *World Neurosurgery* **126**, e937 (2019).
- [24] D. S. Shin, C. P. Carroll, M. Elghareeb, B. L. Hoh, and B. T. Kim, *Journal of Korean Neurosurgical Society* **63**, 137 (2020).
- [25] S. Dandapat, A. Mendez-Ruiz, M. Martínez-Galdámez, J. Macho, S. Derakhshani, G. Foa Torres, V. M. Pereira, A. Arat, A. K. Wakhloo, and S. Ortega-Gutierrez, *Journal of NeuroInterventional Surgery* **13**, 54 (2021).
- [26] P. Bouillot, O. Brina, R. Ouared, K. O. Lovblad, M. Farhat, and V. M. Pereira, *PLoS ONE* **9**, 1 (2014).
- [27] A. A. Dmytriw, K. Phan, J. M. Moore, V. M. Pereira, T. Krings, and A. J. Thomas, *American Journal of Neuroradiology* **40**, 591 (2019).
- [28] G. A. Maragos, A. A. Dmytriw, M. M. Salem, V. M. Tutino, H. Meng, C. Cognard, P. MacHi, T. Krings, and V. M. Pereira, *Neurosurgery* **86**, S21 (2020).
- [29] S. Sindeev, P. G. Arnold, S. Frolov, S. Prothmann, D. Liepsch, A. Balasso, P. Berg, S. Kaczmarz, and J. S. Kirschke, *PLoS ONE* **13**, 1 (2018).
- [30] C. A. Taylor, T. A. Fonte, and J. K. Min, *Journal of the American College of Cardiology* **61**, 2233 (2013), <https://www.jacc.org/doi/pdf/10.1016/j.jacc.2012.11.083>.
- [31] U. Morbiducci, A. M. Kok, B. R. Kwak, P. H. Stone, D. A. Steinman, and J. J. Wentzel, *Thrombosis and Haemostasis* **115**, 484 (2016).
- [32] G. Luraghi, W. Wu, F. De Gaetano, J. F. Rodriguez Matas, G. D. Moggridge, M. Serrani, J. Stasiak, M. L. Costantino, and F. Migliavacca, *Journal of Biomechanics* **58**, 45 (2017).
- [33] M. BEHBAHANI, M. BEHR, M. HORMES, U. STEINSEIFER, D. ARORA, O. CORONADO, and M. PASQUALI, *European Journal of Applied Mathematics* **20**, 363 (2009).
- [34] S. Peschillo, D. Cannizzaro, A. Caporlingua, and P. Missori, *American Journal of Neuroradiology* **37**, 856 (2016).

References

- [35] Z. Kulcsár, E. Houdart, A. Bonafé, G. Parker, J. Millar, A. J. Goddard, S. Renowden, G. Gál, B. Turowski, K. Mitchell, et al., *American Journal of Neuroradiology* **32**, 20 (2011).
- [36] M. Mokin, A. Chinea, C. T. Primiani, Z. Ren, P. Kan, V. M. Srinivasan, R. Hanel, P. Aguilar-Salinas, A. S. Turk, R. D. Turner, et al., *Journal of NeuroInterventional Surgery* **10**, 1074 (2018).
- [37] P. Berg, C. Roloff, O. Beuing, S. Voß, S. I. Sugiyama, N. Aristokleous, A. S. Anayiotos, N. Ashton, A. Revell, N. W. Bressloff, et al., *Journal of Biomechanical Engineering* **137** (2015).
- [38] D. A. Steinman, Y. Hoi, P. Fahy, L. Morris, M. T. Walsh, N. Aristokleous, A. S. Anayiotos, Y. Papaharilaou, A. Arzani, S. C. Shadden, et al., *Journal of Biomechanical Engineering* **135** (2013).
- [39] K. Valen-Sendstad, A. W. Bergersen, Y. Shimogonya, L. Goubergrits, J. Bruening, J. Pallarès, S. Cito, S. Piskin, K. Pekkan, A. J. Geers, et al., *Cardiovascular Engineering and Technology* **9**, 544 (2018).
- [40] P. Berg, S. Voß, S. Saalfeld, G. Janiga, A. W. Bergersen, K. Valen-Sendstad, J. Bruening, L. Goubergrits, A. Spuler, N. M. Cancelliere, et al., *Cardiovascular Engineering and Technology* **9**, 565 (2018).
- [41] S. Voß, O. Beuing, G. Janiga, and P. Berg, *PLoS ONE* **14**, 1 (2019).
- [42] P. Berg, S. Voß, G. Janiga, S. Saalfeld, A. W. Bergersen, K. Valen-Sendstad, J. Bruening, L. Goubergrits, A. Spuler, T. L. Chiu, et al., *International Journal of Computer Assisted Radiology and Surgery* **14**, 1795 (2019).
- [43] R. J. Parkinson, C. S. Eddleman, H. H. Batjer, and B. R. Bendok, *Neurosurgery* **59**, 103 (2006).
- [44] A. Gruber, M. Killer, G. Bavinzski, and B. Richling, *Neurosurgery* **45**, 793 (1999).

References

RÉSUMÉ

Sed tamen haec cum ita tutius observentur, quidam vigore artuum inminuto rogati ad nuptias ubi aurum dextris manibus cavatis offertur, inpigre vel usque Spoletium pergunt. haec nobilium sunt instituta.

MOTS CLÉS

Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method

ABSTRACT

Ego vero sic intellego, Patres conscripti, nos hoc tempore in provinciis decernendis perpetuae pacis habere oportere rationem. Nam quis hoc non sentit omnia alia esse nobis vacua ab omni periculo atque etiam suspicione belli?

KEYWORDS

Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method